

Randomized erasure. Almost any intervention on a neural network will cause its performance to degrade to some extent. Following Elazar et al. [10], we isolate the effect of the concept erasure by comparing it to a control condition in which we orthogonally project onto a *random* linear subspace of the same rank as the cross-covariance matrix. To reduce the variance of our results, we sample a fresh subspace for each minibatch, and erase that subspace at each layer, reporting the cross-entropy loss averaged over subspaces.

Training efficiency. Algorithm 1 avoids redundant computation by caching the layer i hidden states for *every* data point, then using them to run layer $i + 1$. This approach has the downside of requiring a large amount of memory or disk space during training (up to 500GB in our experiments). It’s possible to avoid caching any hidden states and instead recompute them as needed, at the expense of increasing the total compute cost from $O(\ell)$ to $O(\ell^2)$.

6.2 Results

We find strong evidence that autoregressive language models heavily rely on linearly encoded part-of-speech information. While erasing a randomly selected subspace has little to no effect on language modeling performance, scrubbing away part-of-speech information induces a large increase in perplexity across all models (Table 1).

The specific numbers, however, depend on the erasure method used: SAL induces significantly larger increases in perplexity for all models we tested. We take this to mean that SAL inflicts more collateral damage on other useful features in the embedding than LEACE does. In other words, interventions made with LEACE are more *surgical* than those made with prior work; they more closely approximate the ideal of a perfect intervention which only erases the target concept and keeps everything else fixed [40, 15]. If this experiment were conducted with SAL alone, we would have *overestimated* the causal effect of part-of-speech.

7 Limitations and Future Work

Much work remains to be done to validate concept scrubbing. Specifically, we’d like to see experiments that target concepts much narrower than part-of-speech, and use behavioral metrics to determine whether scrubbing changes the network in the ways we’d intuitively expect. If these experiments succeed, an exciting next step would be the incorporation of concept scrubbing into the pretraining and/or finetuning process. This may make it possible to train deep neural networks subject to *conceptual constraints*. It remains to be seen if gradient-based optimizers will be able to “circumvent” such constraints by encoding protected attributes in completely nonlinear ways.

In this work, we focused exclusively on *linear* concept erasure due to its simplicity and tractability. Some authors have proposed nonlinear concept erasure techniques based on kernel methods, but have found that erasure functions fit using one kernel do not generalize well to other kernels [32, 36]. We conjecture that it is intractable to nondestructively edit X so as to prevent a general nonlinear adversary from recovering Z , unless the data generating process for X is known in detail.⁶

A major motivation of concept erasure is that it promises to prevent models from using a concept in a *post hoc*, model-agnostic fashion. But if our concept scrubbing procedure turns out to yield unsatisfactory results in practical use cases, the most promising research direction might then be to improve model-*specific* techniques, such as those that modify the training procedure [8, 9, 14].

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⁶We suspect erasing a concept is at least as hard as extracting it from the original embedding. But in the worst case, information about Z could be encoded *cryptographically* in X , which would be intractable to decode given standard computational complexity assumptions. If the data is generated by a known algorithm, however, it may be possible to efficiently eliminate mutual information between Z and X by simply breaking the links in the causal graph that connect them.

References

- [1] UC Berkeley. The Hilbert space of random variables. Lecture Notes Electrical Engineering 126, 2018. URL <https://inst.eecs.berkeley.edu/~ee126/sp18/projection.pdf>.
- [2] Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. Pythia: A suite for analyzing large language models across training and scaling. *arXiv preprint arXiv:2304.01373*, 2023.
- [3] Tolga Bolukbasi, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai. Man is to computer programmer as woman is to homemaker? De-biasing word embeddings. *Advances in Neural Information Processing Systems*, 29:4349–4357, 2016. URL <https://proceedings.neurips.cc/paper/2016/file/a486cd07e4ac3d270571622f4f316ec5-Paper.pdf>.
- [4] Xilun Chen, Yu Sun, Ben Athiwaratkun, Claire Cardie, and Kilian Weinberger. Adversarial deep averaging networks for cross-lingual sentiment classification. *Transactions of the Association for Computational Linguistics*, 6:557–570, 2018. URL <https://aclanthology.org/Q18-1039>.
- [5] Verna Dankers, Christopher Lucas, and Ivan Titov. Can transformer be too compositional? Analysing idiom processing in neural machine translation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3608–3626, 2022.
- [6] Maria De-Arteaga, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency, FAT* ’19*, page 120–128, New York, NY, USA, 2019. Association for Computing Machinery. ISBN 9781450361255. doi: 10.1145/3287560.3287572. URL <https://doi.org/10.1145/3287560.3287572>.
- [7] Sunipa Dev, Tao Li, Jeff M. Phillips, and Vivek Srikumar. OSCaR: Orthogonal subspace correction and rectification of biases in word embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5034–5050, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.411. URL <https://aclanthology.org/2021.emnlp-main.411>.
- [8] Harrison Edwards and Amos Storkey. Censoring representations with an adversary. In *International Conference in Learning Representations*, pages 1–14, May 2016. URL <https://arxiv.org/abs/1511.05897>.
- [9] Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 11–21, Brussels, Belgium, October–November 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1002. URL <https://aclanthology.org/D18-1002>.
- [10] Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359. URL <https://aclanthology.org/2021.tacl-1.10>.
- [11] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Hatfield Zac Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.
- [12] Thomas S. Ferguson. *Mathematical Statistics*. Academic Press, Cambridge, MA, 1967.

- [13] Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The Pile: An 800GB dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- [14] Atticus Geiger, Zhengxuan Wu, Hanson Lu, Josh Rozner, Elisa Kreiss, Thomas Icard, Noah Goodman, and Christopher Potts. Inducing causal structure for interpretable neural networks. In *International Conference on Machine Learning*, pages 7324–7338. PMLR, 2022.
- [15] Christopher Grimsley, Elijah Mayfield, and Julia R.S. Bursten. Why attention is not explanation: Surgical intervention and causal reasoning about neural models. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 1780–1790, Marseille, France, May 2020. European Language Resources Association. URL <https://aclanthology.org/2020.lrec-1.220>.
- [16] Pantea Haghighatkah, Wouter Meulemans, Bettina Speckmann, Jérôme Urhausen, and Kevin Verbeek. Obstructing classification via projection. In Filippo Bonchi and Simon J. Puglisi, editors, *46th International Symposium on Mathematical Foundations of Computer Science*, Leibniz International Proceedings in Informatics, LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2021.
- [17] Pantea Haghighatkah, Antske Fokkens, Pia Sommerauer, Bettina Speckmann, and Kevin Verbeek. Better hit the nail on the head than beat around the bush: Removing protected attributes with a single projection. pages 8395–8416, December 2022. doi: 10.18653/v1/2022.emnlp-main.575. URL <https://aclanthology.org/2022.emnlp-main.575>.
- [18] Evan Hernandez and Jacob Andreas. The low-dimensional linear geometry of contextualized word representations. In *Proceedings of the 25th Conference on Computational Natural Language Learning*, pages 82–93, 2021.
- [19] Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. spaCy: Industrial-strength Natural Language Processing in Python, 2020.
- [20] Matthäus Kleindessner, Michele Donini, Chris Russell, and Muhammad Bilal Zafar. Efficient fair PCA for fair representation learning. In *International Conference on Artificial Intelligence and Statistics*, pages 5250–5270. PMLR, 2023.
- [21] Abhinav Kumar, Chenhao Tan, and Amit Sharma. Probing classifiers are unreliable for concept removal and detection. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 17994–18008. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/725f5e8036cc08adeba4a7c3bcbcf2c-Paper-Conference.pdf.
- [22] Matt J. Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. *Advances in Neural Information Processing Systems*, 30, 2017.
- [23] Ryan McDonald, Joakim Nivre, Yvonne Quirnbach-Brundage, Yoav Goldberg, Dipanjan Das, Kuzman Ganchev, Keith Hall, Slav Petrov, Hao Zhang, Oscar Täckström, et al. Universal dependency annotation for multilingual parsing. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 92–97, 2013.
- [24] Neel Nanda. Actually, Othello-GPT has a linear emergent world model, Mar 2023. URL <https://neelnanda.io/mechanistic-interpretability/othello>.
- [25] Vassilina Nikoulina, Maxat Tezekbayev, Nuradil Kozhakhmet, Madina Babazhanova, Matthias Gallé, and Zhenisbek Assylbekov. The rediscovery hypothesis: Language models need to meet linguistics. *Journal of Artificial Intelligence Research*, 72:1343–1384, 2021.
- [26] Hamed Nilforoshan, Johann D. Gaebler, Ravi Shroff, and Sharad Goel. Causal conceptions of fairness and their consequences. In *International Conference on Machine Learning*, pages 16848–16887. PMLR, 2022.